with p after E'. Then, since the event of q does not "happen-before" the event of p, there is an execution sequence $E'.p.q$ in which p is dependent with the process $\widehat{e}$ in *backtrack*(E') but need not be in *backtrack*(E'). On the other hand, since $q \in I_{[E']}(p.q)$, the set *backtrack*(E') (together with the initial sleep set at E') is still a source set for the possible continuations after E'.

5.2 Correctness

In this section, we prove that Algorithm 1 is correct in the sense that for each execution sequence E, it explores an execution sequence in $[E.v]_{\simeq}$ for some v. In particular, if E is maximal, then Algorithm 1 explores an execution sequence in $[E]_{\simeq}$. We formalize this statement in Theorem 5.2. In Fig. 5 we give an overview of the structure of its proof.

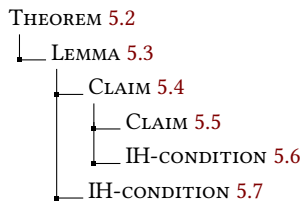

Fig. 5. Structure of correctness proof for Algorithm 1.

We first prove some auxiliary properties of initials and weak initials (Lemma 5.1). We then state Lemma 5.3 (a statement about each call to *Explore*, from which Theorem 5.2 can be derived) and prove it by induction on the order in which states (i.e., execution sequences) are backtracked by the algorithm. A key idea, stated as Claim 5.4, is that when *Explore*(E, Sleep) returns, the set *Sleep* (the *final sleep set*) will be a source set for W after E, where W is the set of suffixes w such that $E.w$ is an execution sequence. To prove Claim 5.4 by contradiction, we assume that there is some $w \in W$ which does not have a weak initial in the final sleep set. Then there is a sequence of form $E.q.w_R$, which (as proven in Claim 5.5) does not have any weak initial in the final sleep set. Now, by the induction hypothesis (which is applicable by IH-condition 5.6) the algorithm will actually explore a sequence equivalent to $E.q.w_R$, in which it will detect a particular race. However, the algorithm guarantees (lines 8 and 9) that the backtrack set at E contains a process which contradicts the initial assumption. Having shown that the final sleep set is a source set for W after E, we can prove Lemma 5.3 by induction (using IH-condition 5.7), thereby also establishing the converse of Theorem 4.4.

We begin by establishing some properties that will be used in the proof:

Lemma 5.1. *The following properties hold:*

(1) If $E \vdash w$, $E \vdash w'$ and $E.w \simeq E.w'$, then
  (a) $I_{[E]}(w) = I_{[E]}(w')$
  (b) $WI_{[E]}(w) = WI_{[E]}(w')$
(2) If $E \vdash w.p$ then:
  (a) $I_{[E]}(w) \subseteq I_{[E]}(w.p)$
  (b) $WI_{[E]}(w.p) \subseteq WI_{[E]}(w)$
(3) $E \vdash q \diamond w \longrightarrow I_{[E.q]}(w) = I_{[E]}(w)$

Proof. We prove each of the cases, in order:

1(*a*). If $p \in I_{[E]}(w)$ then by Definition 4.1(1) there exists w'' such that $E.p.w'' \simeq E.w$. Since $E.w \simeq E.w'$, we then have $E.p.w'' \simeq E.w'$ which by the same definition entails $p \in I_{[E]}(w')$.